

Ryan Murray

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CAREER PROFILE

- Final-year Bachelor of Data Science and Decisions student at UNSW Sydney, majoring in Business Data Science, with strong foundations in machine learning, statistical modelling, and data engineering
- Experienced in building and deploying predictive models and data pipelines in regulated financial services environments, with a focus on delivering actionable insights for business stakeholders
- Skilled at communicating complex analytical findings to both technical and non-technical audiences, with a collaborative, customer-focused approach to problem solving
- Passionate about using data science and AI to improve customer experiences and drive better outcomes for the business and its customers

EDUCATION

Bachelor of Data Science and Decisions | UNSW Sydney

March 2023 – Expected 2026

Business Data Science Major — Business Analytics, Statistical Modelling, Machine Learning, Financial Mathematics, Database Systems

Machine Learning Specialisation | Stanford University & DeepLearning.AI

June 2025

IBM Data Science Professional Certificate

December 2024 – January 2025

WORK EXPERIENCE

Quantitative Risk Analyst Intern | Toyota Finance Australia

Jan – Jul 2026

- Retrained and deployed a production vehicle residual-value model (XGBoost), integrating an external valuation dataset to improve latest-month mean absolute error by ~6% on thin-data EV and new-model vehicles
- Kept all top-50 models by volume within the internal accuracy governance threshold while improving overall backtest mean absolute error by ~3%, validated against the 10% accuracy benchmark
- Built PySpark/SQL data pipelines in Databricks joining 1M+ rows across three data sources to support model retraining, validation, benchmarking, and risk governance workflows, reducing manual processing time
- Designed and documented model monitoring thresholds to detect data drift over time, translating risk governance requirements into clear operational checkpoints aligned with industry standards
- Presented data findings and model performance results to approximately 20 stakeholders across a cross-functional quarterly meeting, communicating technical insights clearly to a mixed audience
- Conducted residual value analysis for emerging electric vehicle models with limited market data, applying benchmark comparisons and producing documentation suitable for internal audit and peer review

Sales Assistant | JB Hi-Fi

October 2024 – Current

- Exceeded individual sales targets of \$10,000+ per shift by applying structured needs analysis to recommend tailored technical solutions for 50+ customers per shift
- Built genuine customer connections through active listening and clear communication, contributing to repeat business and high customer satisfaction scores in a fast-paced, team-based environment

RELEVANT PROJECTS

Portfolio Market-Risk Platform — Automated VaR Monitoring

2026

Python · GARCH · GitHub Actions · Streamlit · Claude API

- Built a self-running market-risk platform that fits GARCH volatility models daily via a scheduled CI pipeline, computes Value-at-Risk and Expected Shortfall via four methods, and validates them with a 500-day walk-forward backtest using Kupiec, Christoffersen, and Basel traffic-light tests
- Engineered for auditability with 32 known-answer unit tests, idempotent versioned artifacts, automated data-quality monitoring with alerting, and an AI layer classifying company announcements into early risk signals

Fraud Detection System — End-to-End Build and Deployment

October 2025

Python · TensorFlow/Keras · AWS · Docker

- Trained an LSTM sequence classifier on 1.85M credit card transactions (0.52% fraud), engineering behavioural features that increased fraud detection precision by 162% and recall by 14%, reaching a test PR-AUC of 0.9865
- Applied cost-based threshold optimisation using industry fraud-cost benchmarks, cutting estimated total fraud cost ~3× versus a standard F1-optimised threshold, and deployed a real-time scoring API in Docker on AWS EC2

Loan Analytics & Risk Modelling

November 2025

Python · Time-Series Forecasting · Logistic Regression · Tableau

- Analysed 30 million loan records to quantify the relationship between interest rates, default risk, and borrower behaviour, building a forecasting model incorporating macroeconomic indicators
- Delivered a 4-dashboard Tableau reporting suite with an interactive risk scenario tool, enabling non-technical stakeholders to explore default scenarios and pricing trade-offs in real time

IoT Device Classifier — Cybersecurity Capstone (Industry-Sponsored, KDDI Corp)

Aug – Oct 2024

Python · Random Forest · Gradient Boosting

- Built a 25-class network device classifier from 19 million records, achieving 86% accuracy and 78% Macro F1, enabling automated device identification at scale without accessing private network content
- Diagnosed and resolved a concept drift issue causing a 25-percentage-point accuracy drop across time periods, lifting cross-month Macro F1 from 62% to 78%, and delivered a formal technical report to the industry sponsor

VOLUNTEERING EXPERIENCE

Yellow Shirts Volunteer Team Leader | Arc @ UNSW

Dec 2024 – Feb 2026

- Progressed from volunteer to team leader across two consecutive orientation programs, managing a 9-person team — coordinating task delegation, resolving challenges, and supporting 50+ first-year students through their transition to university
- Fostered an inclusive and welcoming community for hundreds of incoming students, modelling a positive culture and proactively initiating conversations to support peer connection and belonging

Philanthropy & Sponsorship | UNSW Adventure Society

Mar 2025 – Present

- Raised \$500+ for Save the Children Australia and coordinated fundraising campaigns for Cancer Council Australia, managing outreach and logistics to drive both financial outcomes and community awareness
- Liaised with corporate sponsors to secure support for society events, managing partner relationships and contributing to a sponsorship pipeline that funds ongoing social impact activities

TECHNICAL SKILLS

Programming & Data Analysis: Python (pandas, scikit-learn, TensorFlow/Keras, statsmodels), R, SQL, PySpark, data pipeline design

Machine Learning & AI: Supervised and deep learning models, time-series forecasting, logistic regression, model evaluation, validation, and deployment

Risk & Governance: VaR/ES modelling, model backtesting, model monitoring, data drift detection, credit risk frameworks, model documentation, governance best practices

Cloud & Deployment: AWS, Docker, Databricks, Git, GitHub Actions CI/CD, cloud-native solutions

Data Visualisation & Analytics: Tableau, Power BI, Streamlit, statistical analysis, hypothesis testing, stakeholder reporting